

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

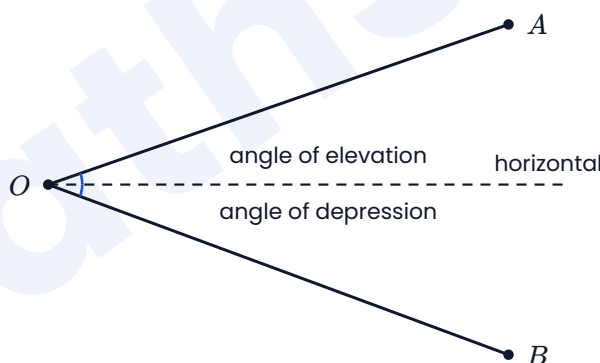
# Chapter 22: Heights and Distances

Topic-wise short notes with solved examples

Trigonometric ratios are used to find heights and distances that cannot be measured directly. Every problem comes down to one or two right-angled triangles.

## 1 Angles of elevation and depression

- The **line of sight** is the line from the observer's eye to the object.
- The **angle of elevation** is the angle between the horizontal and the line of sight when the object is **above** the level of the eye.
- The **angle of depression** is the angle between the horizontal and the line of sight when the object is **below** the level of the eye.
- The angle of depression of  $B$  from  $A$  equals the angle of elevation of  $A$  from  $B$  (alternate angles).



Both angles are measured from the horizontal through the eye O

### REMEMBER

Both angles are measured from the horizontal line through the observer's eye, never from the vertical.

## 2 Which ratio to use

| Known and required sides                       | Ratio  |
|------------------------------------------------|--------|
| Height and horizontal distance                 | $\tan$ |
| Height and slant length (ladder, rope, string) | $\sin$ |
| Horizontal distance and slant length           | $\cos$ |

- Standard values:  $\tan 30^\circ = \frac{1}{\sqrt{3}}$ ,  $\tan 45^\circ = 1$ ,  $\tan 60^\circ = \sqrt{3}$ ,  $\sin 30^\circ = \cos 60^\circ = \frac{1}{2}$ ,  $\sin 60^\circ = \cos 30^\circ = \frac{\sqrt{3}}{2}$ .
- For other angles use the tables. Give the answer to the accuracy asked: nearest metre, or a number of significant figures.

### 3 Problems with one triangle

**Example 3.1** The angle of elevation of the top of a tower from a point 40 m from its foot is  $60^\circ$ . Find the height of the tower, correct to the nearest metre.

**SOLUTION**

$$\tan 60^\circ = \frac{h}{40}, \text{ so } h = 40\sqrt{3} = 40 \times 1.732$$

$h = 69.28$  m, which is 69 m to the nearest metre.

**Example 3.2** A kite is flying at a height of 45 m. Its string makes an angle of  $30^\circ$  with the horizontal. Find the length of the string, assuming it is straight.

**SOLUTION**

$$\sin 30^\circ = \frac{45}{l}, \text{ so } l = \frac{45}{1/2}$$

The string is 90 m long.

**Example 3.3** From the top of a lighthouse 80 m high, the angle of depression of a boat is  $45^\circ$ . Find the distance of the boat from the foot of the lighthouse.

**SOLUTION**

The angle of elevation of the top from the boat is also  $45^\circ$ .

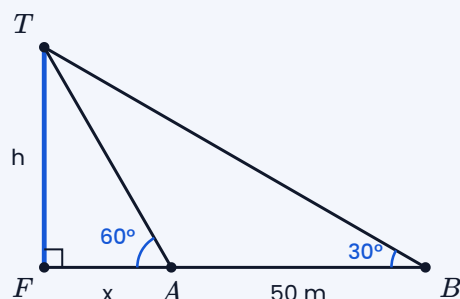
$$\tan 45^\circ = \frac{80}{d}$$

The boat is 80 m from the lighthouse.

### 4 Problems with two triangles

- Mark the common side, call the unknowns  $h$  and  $x$ , write one equation from each triangle and solve the two together.

**Example 4.1** From a point on the ground, the angle of elevation of the top of a tower is  $30^\circ$ . On walking 50 m towards the tower, the angle of elevation becomes  $60^\circ$ . Find the height of the tower.



The tower FT seen from B ( $30^\circ$ ) and from A ( $60^\circ$ )

**SOLUTION**

Let the height be  $h$  and the final distance from the tower be  $x$ .

$$\tan 60^\circ = \frac{h}{x} \text{ gives } h = x\sqrt{3}, \text{ and } \tan 30^\circ = \frac{h}{x+50} \text{ gives } x+50 = \frac{h}{\sqrt{3}}$$

Substituting:  $x+50 = 3x$ , so  $x = 25$  and  $h = 25\sqrt{3}$

**Height** =  $25\sqrt{3} \approx 43.3 \text{ m}$

**Example 4.2** Two ships are on opposite sides of a lighthouse 100 m high, in line with its foot. The angles of depression of the ships from the top are  $30^\circ$  and  $45^\circ$ . Find the distance between the ships.

**SOLUTION**

Ship 1:  $\tan 30^\circ = \frac{100}{d_1}$ , so  $d_1 = 100\sqrt{3} = 173.2 \text{ m}$

Ship 2:  $\tan 45^\circ = \frac{100}{d_2}$ , so  $d_2 = 100 \text{ m}$

**Distance between the ships** =  $173.2 + 100 = 273.2 \text{ m}$

**Example 4.3** From the top of a cliff 60 m high, the angles of depression of the top and the bottom of a tower are  $30^\circ$  and  $60^\circ$ . Find the height of the tower.

**SOLUTION**

Let the horizontal distance be  $d$ . Bottom of the tower:  $\tan 60^\circ = \frac{60}{d}$ , so  $d = \frac{60}{\sqrt{3}} = 20\sqrt{3} \text{ m}$

Top of the tower: the cliff top is  $y$  m above the top of the tower, where  $\tan 30^\circ = \frac{y}{d}$ , so  $y = \frac{20\sqrt{3}}{\sqrt{3}} = 20 \text{ m}$

**Height of the tower** =  $60 - 20 = 40 \text{ m}$

**Example 4.4** A vertical pole and a vertical tower stand on level ground. From the top of the pole, the angle of elevation of the top of the tower is  $60^\circ$  and the angle of depression of its foot is  $30^\circ$ . If the tower is 50 m high, find the height of the pole.

**SOLUTION**

Let the pole be  $h$  m high and the distance be  $d$ . Depression of the foot:  $\tan 30^\circ = \frac{h}{d}$ , so  $d = h\sqrt{3}$ .

Elevation of the top:  $\tan 60^\circ = \frac{50 - h}{d}$ , so  $50 - h = d\sqrt{3} = 3h$

$h = 12.5$  m

## 5 Steps for every problem

- Draw a clear figure and label all points.
- Mark the given angles from the horizontal.
- Write a trigonometric ratio for each right-angled triangle.
- Solve, then round off the final answer as asked.

## 6 Common mistakes

- Marking the angle of depression at the object instead of at the observer.
- When two objects are on **opposite** sides, add the distances; on the **same** side, subtract.
- Rounding  $\sqrt{3}$  too early.
- Forgetting the observer's height when it is given.